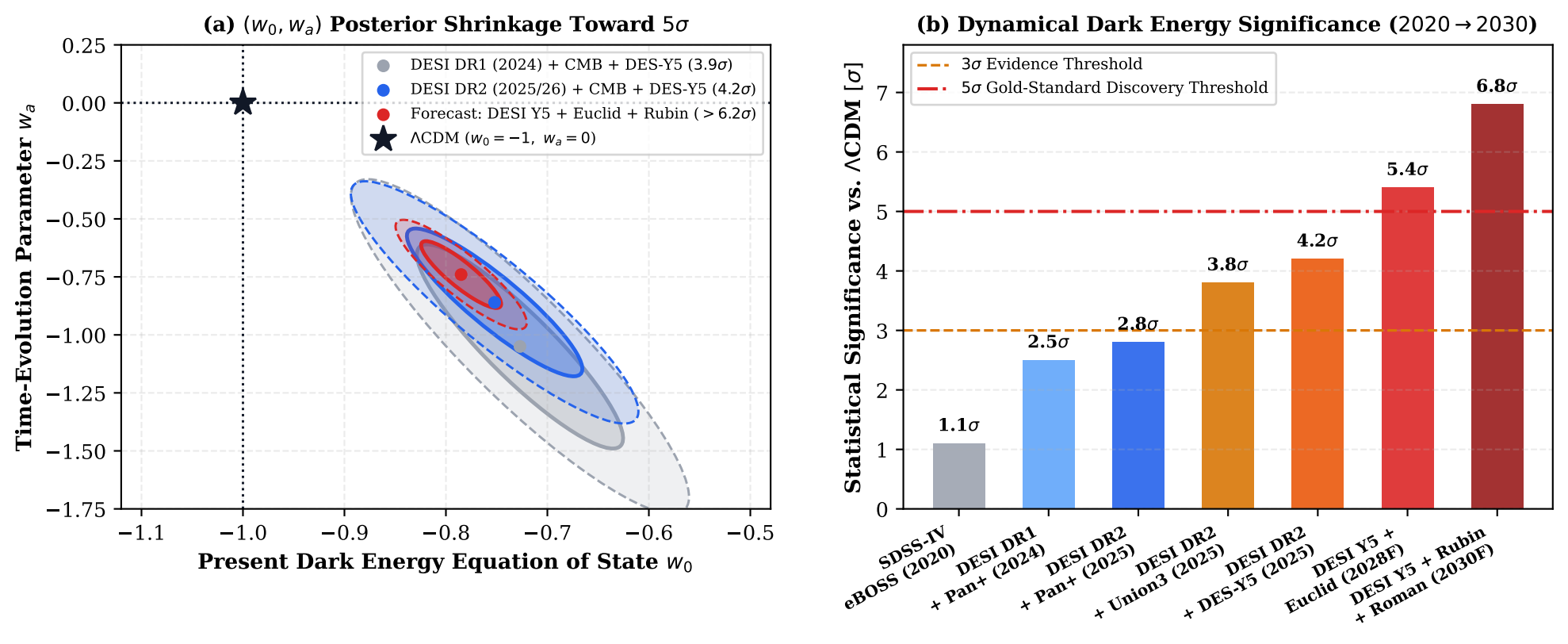
(a) (w_0, w_a) Posterior Shrinkage Toward 5σ
DESI DR1 (2024) + CMB + DES-Y5 (3.9σ)
DESI DR2 (2025/26) + CMB + DES-Y5 (4.2σ)
Forecast: DESI Y5 + Euclid + Rubin ($> 6.2\sigma$)
ΛCDM ($w_0 = -1,\ w_a = 0$)
Time-Evolution Parameter w_a
Present Dark Energy Equation of State w_0
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−1.1
−1.0
−0.9
−0.8
−0.7
−0.6
−0.5
(b) Dynamical Dark Energy Significance (2020 → 2030)
3σ Evidence Threshold
5σ Gold-Standard Discovery Threshold
Statistical Significance vs. ΛCDM [σ]
0
1
2
3
4
5
6
7
1.1σ
2.5σ
2.8σ
3.8σ
4.2σ
5.4σ
6.8σ
SDSS-IV eBOSS (2020)
DESI DR1 + Pan+ (2024)
DESI DR2 + Pan+ (2025)
DESI DR2 + Union3 (2025)
DESI DR2 + DES-Y5 (2025)
DESI Y5 + Euclid (2028F)
DESI Y5 + Rubin + Roman (2030F)